

CAMERON FALLS, ON, Canada

WMO: 712910

Lat: 49.154N

Lon: 88.345W

Elev: 233

StdP: 98.56

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-26.9	-32.5	0.2	-29.1	-30.0	0.2	-26.4	8.4	-14.1	7.6	-13.6	1.0	20	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	27.7	19.3	26.0	18.3	24.4	17.3	21.1	25.6	19.7	23.6	18.6	22.4	3.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.4	14.6	23.6	18.2	13.5	21.9	17.1	12.6	20.3	62.2	25.7	57.3	23.5	53.5	22.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	6.1	-32.5	31.2	2.8	1.8	-34.5	32.5	-36.2	33.5	-37.8	34.5	-39.8	35.8		
(5)			-32.6	23.1	2.8	1.6	-34.6	24.3	-36.3	25.2	-37.8	26.1	-39.9	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.8	-14.0	-12.9	-6.0	1.6	8.2	14.1	17.6	17.1	12.6	5.6	-1.8	-10.1		
	DBStd	12.24	7.29	6.48	6.73	4.56	3.97	3.05	2.86	3.10	4.10	4.28	5.73	6.91		
	HDD10.0	3371	744	641	496	253	84	4	0	0	20	150	354	624		
	HDD18.3	5744	1003	874	754	503	314	129	49	61	176	394	604	883		
	CDD10.0	724	0	0	0	1	28	128	234	222	98	14	0	0		
	CDD18.3	56	0	0	0	0	0	3	25	23	4	1	0	0		
	CDH23.3	589	0	0	0	0	15	67	282	194	30	1	0	0		
	CDH26.7	104	0	0	0	0	1	9	56	33	4	1	0	0		
(14) Wind	WSAvg	2.3	2.2	2.4	2.6	2.8	2.7	2.2	2.1	2.0	2.2	2.3	2.3	2.1		
(15) Precipitation	PrecAvg	867	54	33	47	45	67	95	90	104	99	91	78	64		
	PrecMax	1243	118	82	152	103	136	222	172	202	217	240	158	137		
	PrecMin	607	11	6	8	7	9	36	39	40	44	30	26	25		
	PrecStd	150	28	19	30	25	30	48	36	40	45	47	33	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	4.4	13.3	17.7	25.5	27.9	30.5	29.9	26.5	20.7	12.6	4.5		
		MCWB	1.6	1.6	7.5	9.0	14.7	18.5	20.5	20.7	18.5	14.4	9.4	3.4		
	2%	DB	1.2	1.8	8.2	13.9	22.2	25.1	28.3	27.2	23.5	17.3	9.5	2.9		
		MCWB	0.5	0.4	4.6	6.8	13.3	16.3	19.3	19.5	17.6	12.6	7.4	2.0		
	5%	DB	-0.1	-0.3	4.9	11.6	19.3	23.3	26.4	25.5	21.2	14.6	7.6	1.5		
		MCWB	-0.6	-1.5	1.9	5.6	11.8	15.6	18.7	18.5	16.4	11.4	5.9	0.8		
	10%	DB	-2.5	-2.5	3.0	9.4	16.6	21.5	24.6	23.7	19.1	12.5	5.9	0.1		
		MCWB	-3.2	-3.6	1.0	4.5	10.5	14.5	17.8	17.7	15.6	10.2	4.6	-0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.0	2.4	8.6	9.9	17.0	20.2	22.6	23.0	20.4	16.0	10.1	3.7		
		MCDB	2.5	3.5	11.4	14.9	20.9	24.4	27.7	28.5	24.0	18.3	12.0	4.2		
	2%	WB	0.7	0.5	4.7	7.7	14.5	18.0	21.0	21.2	18.7	13.6	7.8	2.2		
		MCDB	1.1	1.7	7.4	12.6	19.2	22.5	26.3	25.4	21.6	15.9	9.1	2.7		
	5%	WB	-0.7	-1.5	2.6	6.2	12.9	16.6	19.5	19.9	17.4	11.9	6.1	0.9		
		MCDB	-0.1	-0.5	4.4	10.2	17.9	21.3	24.2	23.4	19.9	14.0	7.2	1.3		
	10%	WB	-3.2	-3.6	1.1	4.8	11.4	15.5	18.4	18.7	16.3	10.3	4.6	-0.5		
		MCDB	-2.5	-2.6	2.7	8.7	15.6	19.6	22.9	22.1	18.5	12.4	5.9	0.2		
(35) Mean Daily Temperature Range	MDBR		9.7	11.0	11.3	10.6	11.6	11.9	11.6	10.9	9.9	7.5	6.7	7.9		
	5% DB	MCDBR	8.8	11.2	12.2	15.1	17.6	16.2	15.1	14.1	12.4	11.9	8.6	7.7		
		MCWBR	8.2	9.4	8.5	8.7	9.2	8.2	7.4	7.3	7.3	7.8	6.2	7.1		
	5% WB	MCDBR	8.5	10.5	10.4	13.5	13.8	12.9	13.3	12.2	10.2	10.4	7.9	7.0		
		MCWBR	8.0	9.2	7.3	8.4	8.0	7.4	7.3	7.0	6.5	7.4	6.1	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.269	0.301	0.342	0.354	0.372	0.393	0.368	0.354	0.312	0.284	0.264		
	taud		2.351	2.318	2.345	2.376	2.428	2.402	2.381	2.447	2.483	2.568	2.496	2.361		
	Ebn at Noon		818	890	915	906	906	888	862	870	849	837	783	751		
	Edh at Noon		66	87	101	110	109	113	114	102	89	69	58	58		
(44) All-Sky Solar Radiation	RadAvg		1.19	2.31	3.61	4.77	5.33	5.72	5.73	4.89	3.48	2.03	1.05	0.85		
	RadStd		0.15	0.21	0.32	0.47	0.31	0.41	0.37	0.40	0.27	0.28	0.11	0.11		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page